

MISSION FIELD, MT, USA

WMO: 726798

Lat: 45.698N

Lon: 110.441W

Elev: 1415

StdP: 85.44

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24150

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.8	-19.3	-26.6	0.4	-22.6	-23.0	0.6	-18.7	22.7	2.1	21.3	1.3	4.2	320	0.782

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	33.9	15.5	32.2	15.1	30.4	14.7	17.3	29.0	16.5	28.1	15.7	27.2	5.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	11.6	20.1	12.4	10.7	19.4	11.3	9.9	18.9	53.8	28.8	51.1	27.9	48.6	27.2	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.7	16.3	14.6	DB	-28.6	37.0	3.8	1.1	-31.4	37.8	-33.6	38.5	-35.7	39.1	-38.4	40.0	(4)
(5)			WB	-28.8	19.1	3.7	1.1	-31.5	20.0	-33.7	20.6	-35.8	21.3	-38.5	22.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	-1.6	-2.3	2.5	6.0	10.4	15.4	20.7	19.3	14.4	7.8	2.0	-2.6	(6)	
	DBStd	9.80	6.85	7.42	6.56	4.87	4.44	3.85	2.89	3.19	4.56	5.79	7.33	7.19	(7)	
	HDD10.0	1897	360	346	241	137	49	5	0	0	13	107	249	391	(8)	
	HDD18.3	4048	618	579	492	372	246	102	11	27	133	329	490	649	(9)	
	CDD10.0	1065	1	1	7	15	63	167	332	289	144	38	9	0	(10)	
	CDD18.3	173	0	0	0	0	1	14	85	58	14	1	0	0	(11)	
	CDH23.3	3544	0	0	0	6	75	356	1489	1179	412	26	0	0	(12)	
	CDH26.7	1512	0	0	0	0	12	122	705	524	145	4	0	0	(13)	
(14) Wind	WSAvg	6.3	9.0	7.6	7.1	5.9	5.2	4.7	4.2	4.4	4.8	6.4	8.1	8.6	(14)	
(15) Precipitation	PrecAvg	392	14	9	21	35	66	65	33	37	44	37	21	10	(15)	
	PrecMax	581	27	24	56	82	132	135	82	69	80	88	61	19	(16)	
	PrecMin	271	2	0	8	13	14	14	4	6	1	5	2	0	(17)	
	PrecStd	83	8	7	12	20	29	33	23	22	24	21	18	5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.4	20.4	24.0	28.3	33.0	36.6	35.6	32.7	26.3	19.4	12.6	(19)	
		MCWB	5.1	5.1	8.1	10.4	13.4	15.9	16.7	15.6	14.5	11.7	8.6	4.9	(20)	
	2%	DB	10.3	11.6	17.0	20.9	25.5	30.1	34.5	33.5	30.5	22.8	16.4	10.3	(21)	
		MCWB	3.4	4.2	6.6	9.3	12.8	15.4	15.9	15.1	13.7	10.4	7.4	4.0	(22)	
	5%	DB	8.7	9.6	14.2	18.1	22.9	27.4	32.8	31.7	28.2	19.9	14.1	8.5	(23)	
		MCWB	2.6	3.1	5.5	8.0	11.8	14.8	15.4	14.7	12.9	9.2	6.1	2.7	(24)	
	10%	DB	7.0	7.6	11.8	15.2	20.2	24.9	31.0	29.9	25.5	17.3	11.9	6.8	(25)	
		MCWB	1.6	1.8	4.5	7.0	10.9	14.0	15.3	14.3	12.1	8.2	4.8	1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.6	6.0	8.5	11.6	15.4	18.0	18.9	17.7	15.8	12.8	9.7	5.7	(27)	
		MCDB	11.9	12.4	18.7	21.6	24.3	28.5	30.0	30.9	27.5	24.3	18.1	11.5	(28)	
	2%	WB	4.1	4.7	7.1	9.9	13.8	16.7	17.7	16.4	14.6	11.1	7.9	4.5	(29)	
		MCDB	9.4	10.9	15.9	19.2	22.7	26.8	29.2	28.8	27.1	20.9	15.7	9.7	(30)	
	5%	WB	2.9	3.4	5.9	8.5	12.6	15.6	16.8	15.6	13.6	10.0	6.5	3.1	(31)	
		MCDB	8.1	9.1	13.3	17.3	21.6	25.2	28.7	28.0	25.7	18.4	13.4	8.1	(32)	
	10%	WB	1.8	2.0	4.8	7.2	11.4	14.6	16.1	14.9	12.5	8.7	5.1	1.8	(33)	
		MCDB	6.5	7.2	11.3	14.7	19.3	23.8	28.0	27.2	23.8	16.3	11.3	6.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	10.3	11.7	12.6	13.7	15.3	19.0	19.0	16.6	12.8	10.1	8.8	(35)	
		MCDBR	9.7	12.1	14.7	18.3	19.3	20.8	22.1	22.7	22.3	17.8	12.3	9.0	(36)	
	5% WB	MCWBR	5.8	7.1	7.5	9.1	8.9	8.7	7.6	8.2	9.1	8.5	6.5	5.4	(37)	
		MCWBR	9.5	11.5	13.8	17.4	17.8	18.3	19.2	19.3	19.6	16.4	12.3	8.9	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.245	0.252	0.271	0.310	0.320	0.327	0.335	0.346	0.319	0.287	0.259	0.241	(40)	
		taud	2.428	2.418	2.483	2.415	2.471	2.496	2.498	2.432	2.501	2.560	2.537	2.469	(41)	
	Edn at Noon	Edn at Noon	888	947	972	951	947	938	925	902	904	894	867	858	(42)	
		Edh at Noon	68	84	92	109	107	104	103	106	90	73	61	59	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.61	2.59	3.76	4.79	5.49	6.57	7.06	6.02	4.52	2.82	1.73	1.28	(44)	
	RadStd	RadStd	0.14	0.18	0.25	0.32	0.54	0.52	0.33	0.33	0.38	0.22	0.16	0.09	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page